Trainings

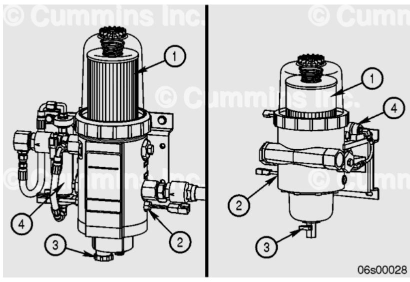
General Information

Note : This procedure applies to Stage 1 fuel filters on engines with electronically actuated injector only.

Use the following procedure for mechanically actuated injectors. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-015-tr.html)

Use the following procedure for Stage 2 fuel filters. Refer to Procedure 006-076 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-076-tr.html)

Industrial and Power Generation Applications



- The Stage 1 filter mounting location is original equipment manufacturer (OEM)-dependent, provided Cummins Inc. application guidelines are met. Contact a Cummins® Authorized Repair Location for further information.
- Industrial and Power Generation applications are explained in the following steps.

Stage 1 filters are mounted off-engine. The fuel priming pump is mounted with the Stage 1 filter assembly.

The FH234 series filter is shown on the left side of the illustration and the FH239 series on the right. The filter consists of the following components:

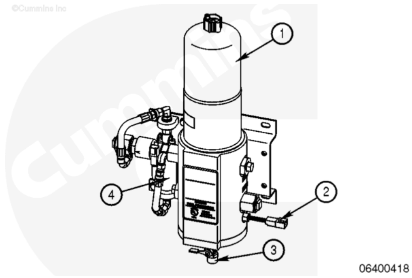
1. Stage 1 filter
2. Water in fuel sensor
3. Drain valve
4. Lift pump.

Note : The following illustrations show the FH234 series, but the FH239 series steps are the same.

Marine Non-Classed Applications - Spin-On Filters

- The Stage 1 filter mounting location is OEM dependent, provided Cummins Inc. application guidelines are met.

Contact a Cummins® Authorized Repair Location for further information.
- The Marine application engines are equipped with two different filter systems. They are identified as Classed and Non-Classed. Classed applications enable the engine to remain in operation while the filters are changed. Non-Classed applications require the engine to be shut down for service.
- Non-Classed applications are explained in the following steps.

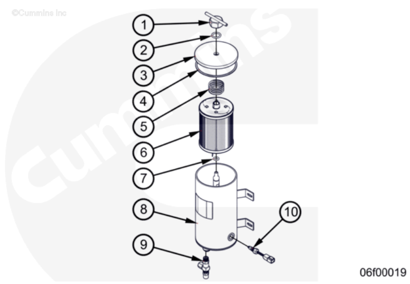


The Marine Stage 1 filter is mounted off-engine. The Non-Classed filter arrangement is equipped with a single Sea Pro® filter, with the lift pump mounted to the filter bracket.

The Sea Pro® FH234 single filter consists of the following:

1. Stage 1 filter (7 micron filter)
2. Water in fuel sensor
3. Drain valve
4. Lift pump.

Marine Non-Classed Applications - Cartridge Filters



- The Stage 1 filter mounting location is OEM dependent, provided Cummins Inc. application guidelines are met. Contact a Cummins® Authorized Repair Location for further information.
- This Marine application engines are equipped with two different filter systems. They are identified as Classed and Non-Classed. Classed applications enable the engine to remain in operation while the filters are changed. Non-Classed applications require the engine to be shutdown for service.
- Non-Classed applications are explained in the following steps.

The Marine Stage 1 filter is mounted off-engine. The Non-Classed filter arrangement is equipped with a single Sea Pro® filter, with the lift pump mounted on-engine.

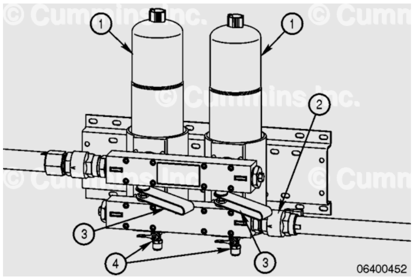
The Sea Pro® FH240 single filter consists of the following:

1. Nut
2. Nut Seal
3. Service Cap
4. Service Cap Seal
5. Spring

6. Stage 1 Filter Element and Seals (Includes 2, 4, and 7)
7. Stand Pipe Gasket
8. Filter Housing
9. Water in fuel Sensor
10. Drain Valve.

For Classed applications, the lift pump is mounted on-engine, and **not** on the Stage 1 filter head assembly.

Marine Classed Applications - Spin-On Filters



- The Stage 1 filter mounting location is OEM dependent, provided Cummins Inc. application guidelines are met. Contact a Cummins® Authorized Repair Location for further information.
- This procedure explains Classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut off, follow the Non-Classed procedure.
- Classed applications are explained in the following steps.

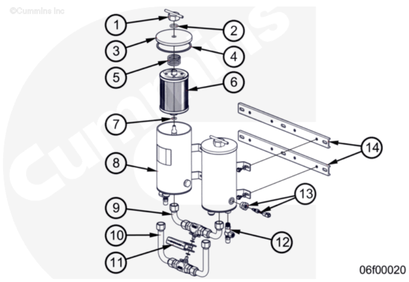
Marine Classed applications use a duplex Stage 1 filter that utilizes valves to allow for changing of the filters while the engine is in operation at reduced capability or at idle, with the lift pump mounted to the filter bracket.

The valves are located between the inlet and outlet fuel manifolds on the filter assembly.

The Sea Pro® FH234 duplex filter consists of the following:

1. Stage 1 filters (7 micron filter)
2. Water in fuel sensor
3. Shutoff valves
4. Drain valves.

Marine Classed Applications - Cartridge Filters



- The Stage 1 filter mounting location is OEM dependent, provided Cummins Inc. application guidelines are met. Contact a Cummins® Authorized Repair Location for further information.
- This procedure explains Classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut off, follow the Non-Classed procedure.
- Classed applications are explained in the following steps.

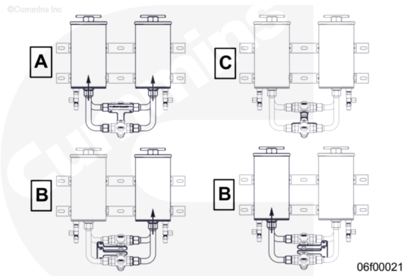
Marine Classed applications use a duplex Stage 1 filter that utilizes valves to allow for changing of the filters while the engine is in operation at reduced capability or at idle, with the lift pump mounted on engine.

The valve is located between the inlet and outlet fuel manifolds on the filter assembly

The Sea Pro® FH240 duplex filter consists of the following:

1. Nut
2. Nut Seal
3. Service Cap
4. Service Cap Seal
5. Spring
6. Stage 1 Filter Element and Seals (Includes 2, 4, and 7)
7. Stand Pipe Gasket
8. Filter Housing
9. Fuel Out Manifold
10. Fuel In Manifold
11. Handle
12. Drain Valve
13. Water in fuel Sensor
14. Mounting Bracket.

Valve Operation - Cartridge Filters



A. For dual unit flow, the valve handle should be facing upward.
B. For single unit flow, rotate the valve handle either left or right 90 degrees depending on which unit is to be used.
C. To shut off flow, rotate the valve handle 180 degrees from the dual flow position.

Preparatory Steps

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe

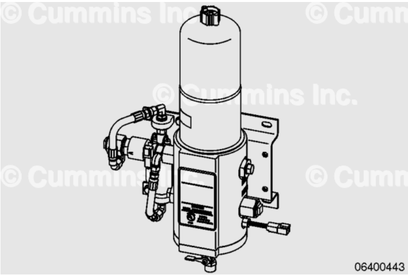
personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

Industrial and Power Generation Applications

- The Stage 1 filtration for the Industrial and Power Generation application QSK19 Modular Common Rail System (MCRS) engine consists of a single Industrial Pro™ filter head. The Stage 1 filtration **must** be used during normal operation.
- Shut the engine down.
- Close the fuel supply shutoff valve.
- Drain the filter head of fuel.

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

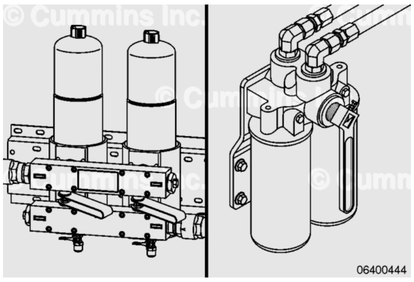


Marine Non-Classed Applications - Spin-On and Cartridge Filters

- The Stage 1 filtration for the Non-Classed Marine application QSK19 MCRS engine consists of a single Sea Pro® filter head. The Stage 1 filtration **must** be used during normal engine operation.
- Shut the engine down.
- Close the fuel supply shutoff valve.
- Drain the filter head of fuel.

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



Note : **This procedure explains Classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut down, follow the Non-Classed procedure.**

Marine Classed Applications - Spin-On and Cartridge Filters

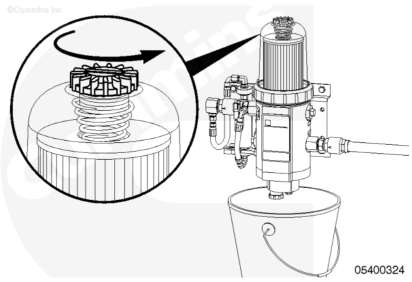
- The Stage 1 filtration for the Classed Marine application QSK19 MCRS engine consists of a duplex Sea Pro® filter head. At least one of the filter elements **must** be used during normal engine operation.
- Drain the fuel from the filter head.

With the engine operating at idle, move the selected valve handle to the OFF position (depending on which Stage 1 filter is to be changed.)

Remove

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



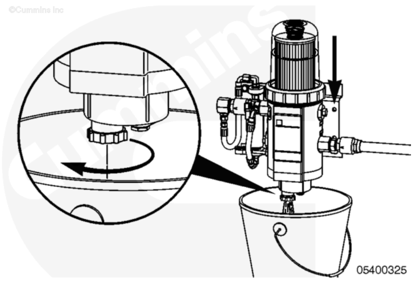
⚠ CAUTION ⚠

Do not allow fuel to drain onto the ground. Drained fuel must be collected and disposed of in accordance with local environmental regulations.

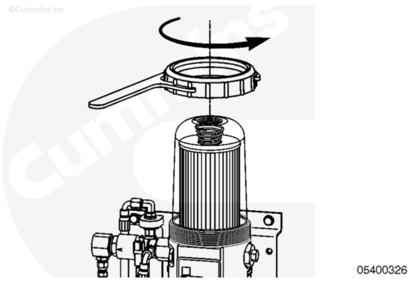
Industrial and Power Generation Applications

- Fuel **must** be drained from filter head prior to removing filter element.
- Place a suitable container under the Stage 1 fuel filter to be replaced.
- Use collar/vent cap wrench, Part Number 3944451S, to open vent cap to break the air lock in the filter.

Open the fuel drain valve and allow the fuel level to drain to a point below the collar. Close the drain valve.



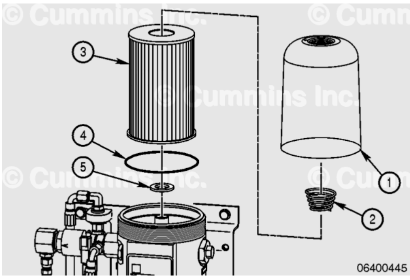
Use collar/vent cap wrench, Part Number 3944451S, to loosen collar.



Remove clear cover (1), filter spring (2), fuel filter element (3), and o-ring (4).

Remove sealing grommet (5) from the filter spud.

Discard o-ring (4) and sealing grommet (5).

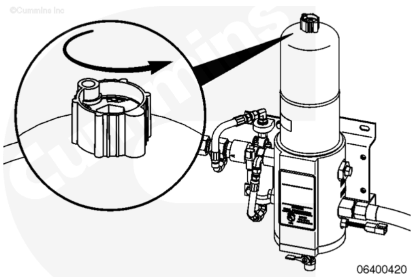


⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.

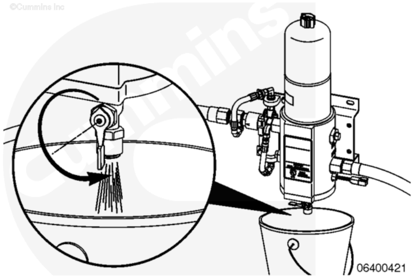


Marine Non-Classed Applications - Spin-On Filters

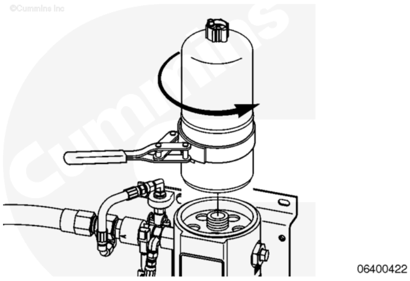
- The keyswitch **must** remain in the OFF position while draining the Stage 1 filter. This will prevent excessive air intake into the fuel system.
- Place a suitable container under the Stage 1 fuel filter to be replaced.
- Open vent cap to break the air lock in the filter.

Open fuel drain valve and allow filter to drain.

Close drain valve when drainage slows to a trickle.



Remove Stage 1 fuel filter with filter wrench, Part Number 3400157, or equivalent.



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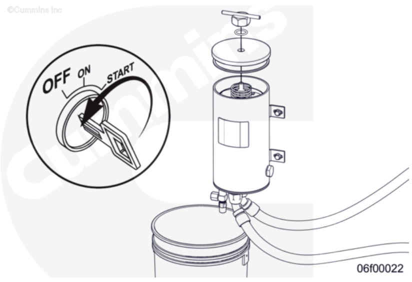


⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.

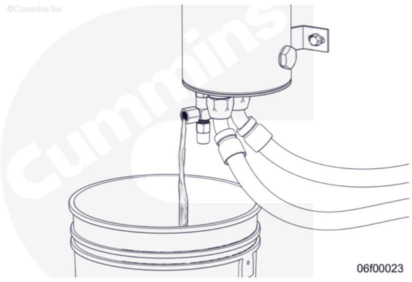
Note : This procedure explains Classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut down, follow the Non-Classed procedure.

Marine Non-Classed Applications - Cartridge Filters

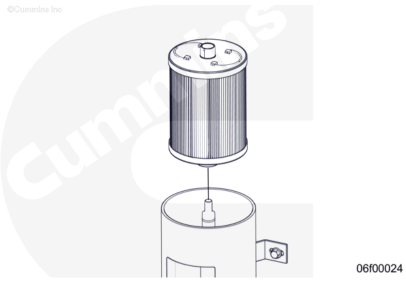
- The keyswitch **must** remain in the OFF position while draining the Stage 1 filter. This will prevent excessive air intake into the fuel system.
- Place a suitable container under the Stage 1 fuel filter to be replaced.
- Remove service cap to break the air lock in the filter.

Open the fuel drain valve and allow filter to drain.

Close drain valve when drainage slows to a trickle.



Remove the Stage 1 fuel filter.



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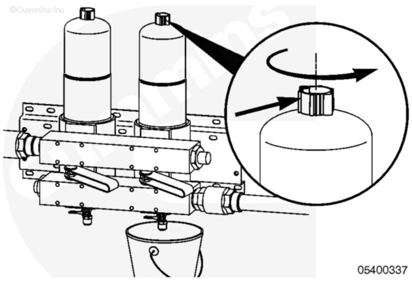


⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.

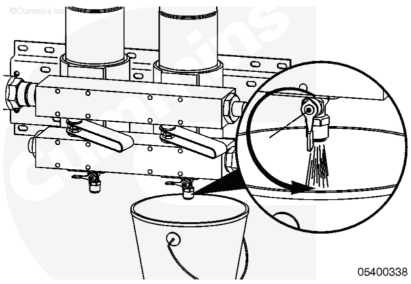
Note : **This procedure explains classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut down, follow the Non-Classed procedure.**

Marine Classed Applications - Spin-On Filters

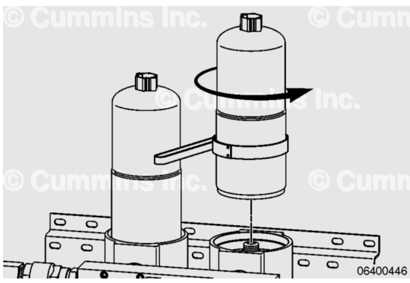
- After filter element has been removed, make sure the selector valve remains in the OFF position.
- Place a suitable container under the Stage 1 fuel filter to be replaced.
- Open the vent cap to break the air lock in the filter.

Open fuel drain valve and allow filter to drain.

Close drain valve when drainage slows to a trickle.

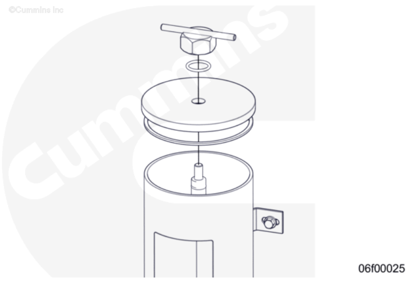


Remove Stage 1 fuel filter with filter wrench, Part Number 3400157, or equivalent.



WARNING

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



CAUTION

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.

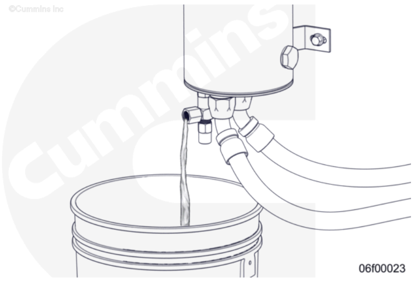
Note : **This procedure explains Classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut down, follow the Non-Classed procedure.**

Marine Classed Applications - Cartridge Filters

- Place the selector valve in OFF position.
- Place a suitable container under the Stage 1 fuel filter to be replaced.
- Remove the service cap to break the air lock in the filter.

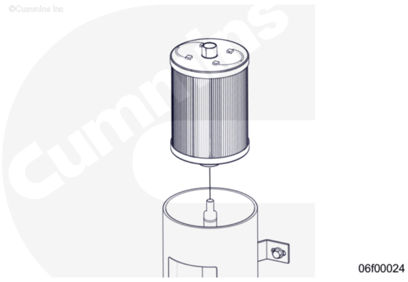
Open fuel drain valve and allow filter to drain.

Close drain valve when drainage slows to a trickle.



Remove the Stage 1 fuel filter.

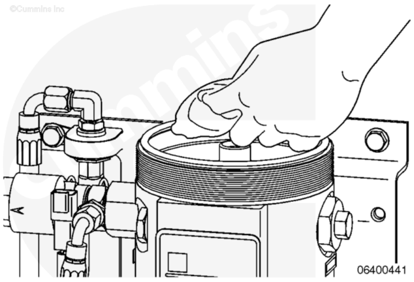
After filter element has been removed, make sure selector valve remains in the OFF position.



Clean

Note : **Shown is the Industrial application filter. Although different, the procedure remains the same for both Industrial and Marine applications.**

Check the sealing surface of the filter body. Wipe surface with a clean, lint-free cloth to remove any debris or dirt.



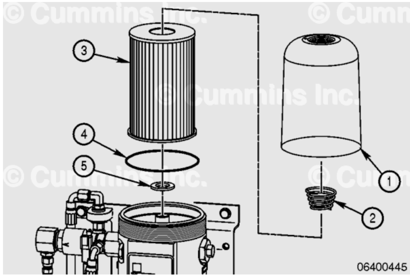
Install

Note : Verify used o-ring has been removed and discarded before installing new o-ring.

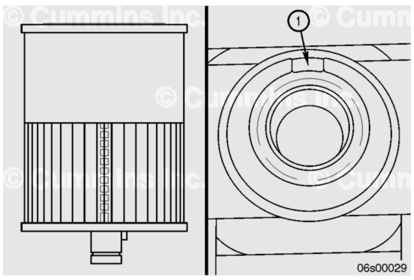
Industrial and Power Generation Applications

- Stage 1 filter element replacement includes the appropriate o-ring (4) and sealing grommet (5).
- The o-ring and grommet **must** be replaced with the filter element to make sure of correct operation.

On the Stage 1 filter, install a new pre-lubricated o-ring (4), filter element (3) (supplied with a sealing grommet (5) inserted into the filter element), filter spring (2), and clear cover (1).

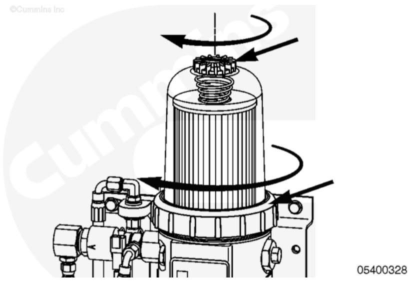


The FH239 series element has a key that **must** match the keyway (1) in the filter head.



Install vent cap and collar onto clear cover and hand-tighten **only**.

Do **not** use tools to tighten collar.

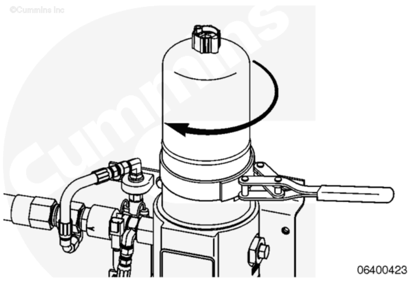


Marine Application - Spin-On Filters

- The Stage 1 filter element replacement includes the appropriate o-ring.
- The o-ring **must** be replaced with the filter element for correct operation.

Install Stage 1 filter on the filter head. Turn filter until the pre-lubricated o-ring touches the surface of the filter head.

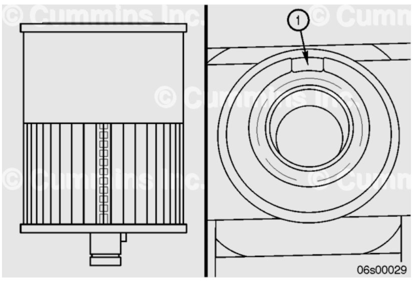
Tighten filter an additional ½ to ¾ of a turn.



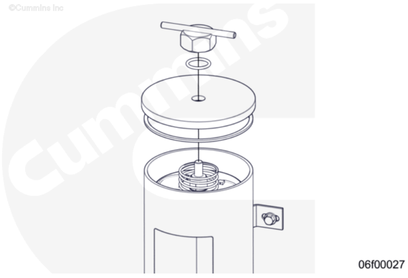
Marine Applications - Cartridge Filters

- The Stage 1 filter element replacement includes the appropriate o-ring.
- The o-ring **must** be replaced with the filter element to make sure of correct operation.

Install a new pre-lubricated o-ring on the filter element. Insert the filter element into the filter housing. The FH240 series element has a key that **must** match the keyway (1) in the filter head.



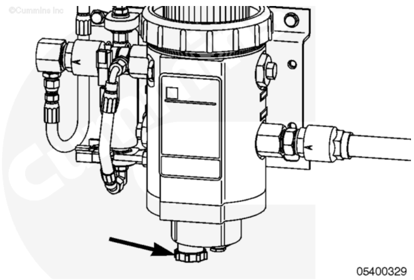
Place service cap on top of filter housing with replacement service cap seal, making sure cap is centered on housing. Install knob and hand-tighten.



Prime

⚠ CAUTION ⚠

Do not spill or drain fuel onto the ground when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements onto the ground. The fuel must be disposed of in accordance with local environmental regulations.



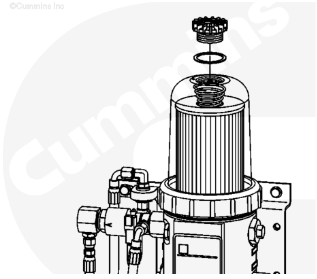
Industrial and Power Generation Applications

- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.
- Procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows.

Verify drain valve on the Stage 1 filter is closed at the base of the filter.

Close fuel supply shutoff valve.

Remove vent cap from the top of the filter.



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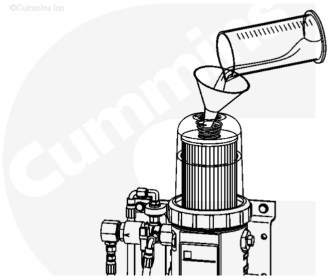
⚠ CAUTION ⚠

Do not spill or drain fuel onto the ground when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements onto the ground. The fuel must be disposed of in accordance with local environmental regulations.



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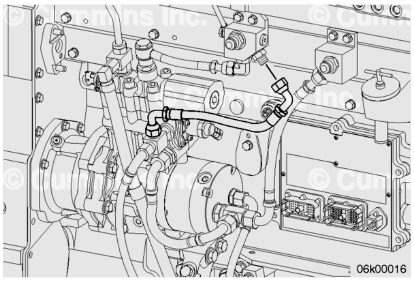
Fill Stage 1 fuel filter with clean diesel fuel.

Install the vent cap.

Tighten vent cap by hand **only**.

Open fuel supply shutoff valve.

Remove air bleed line from drain manifold block and place it into a collection container.

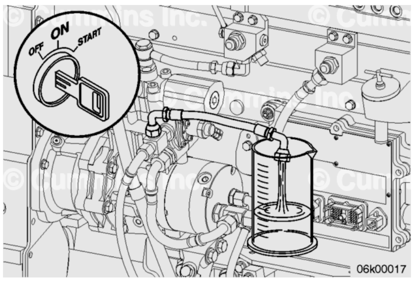


Turn keyswitch ON and observe fuel exiting the air bleed line.

When steady stream of fuel is observed, turn keyswitch OFF.

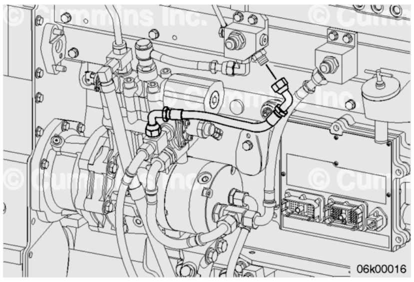
If priming pump stops operating before a steady stream of fuel is observed, turn keyswitch OFF, wait 30 seconds, and then turn keyswitch back ON.

Repeat as necessary to observe a steady stream of fuel.



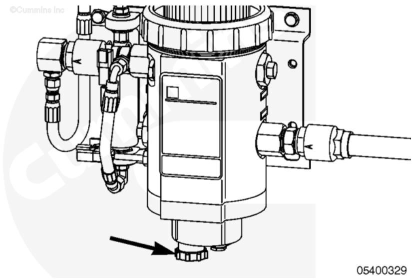
Install an air bleed line to drain the manifold block.

Torque Value: 45 n•m [33 ft-lb]



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.



Marine Non-Classed Applications - Spin-On Filters

- Stage 1 and Stage 2 filters must be primed prior to starting the engine.
- Procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows.

Verify drain valve is closed at base of the Stage 1 filter.

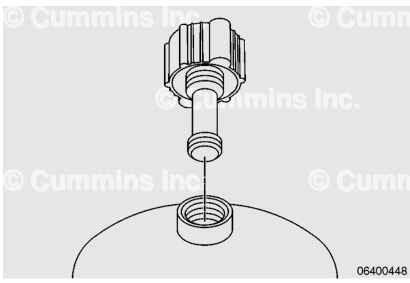
Close fuel supply shutoff valve.

Unscrew threads of the vent cap.

Lift vent cap until the retainer shaft catches.

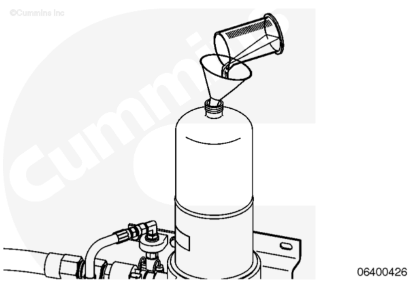
Unscrew retainer shaft from the threaded boss in the top of filter can.

Remove vent cap from top of the can.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.



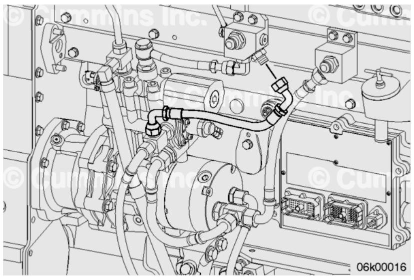
Fill Stage 1 fuel filter with clean diesel fuel.

Install vent cap.

Tighten vent cap by hand **only**.

Open fuel supply shutoff valve.

Remove air bleed line from the drain manifold block and place it into a collection container.

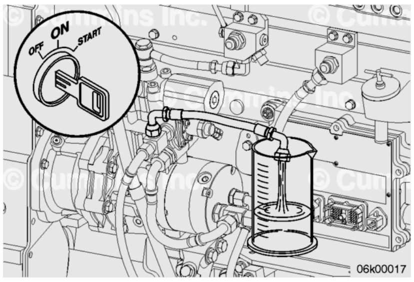


Turn keyswitch ON and observe fuel exiting air bleed line.

When a steady stream of fuel is observed, turn keyswitch OFF.

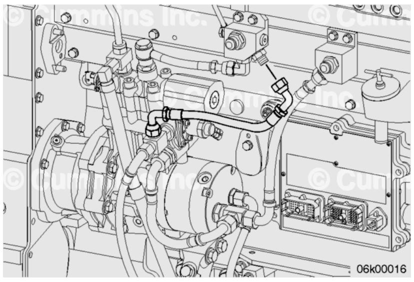
If priming pump stops operating before a steady stream of fuel is observed, turn keyswitch OFF, wait 30 seconds, and then turn keyswitch back ON.

Repeat as necessary to observe a steady stream of fuel.



Install an air bleed line to drain the manifold block.

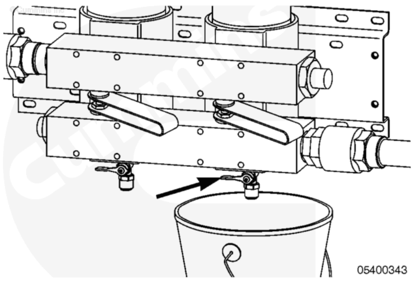
Torque Value: 45 n•m [33 ft-lb]



Note : This procedure explains Classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut down, follow the Non-Classed procedure.

Marine Classed Applications - Spin-On Filters

- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.
- Procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows.

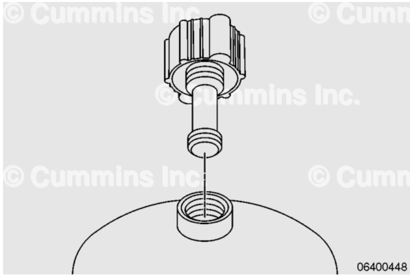


Unscrew threads of the vent cap.

Lift vent cap until the retainer shaft catches.

Unscrew retainer shaft from the threaded boss in top of filter can.

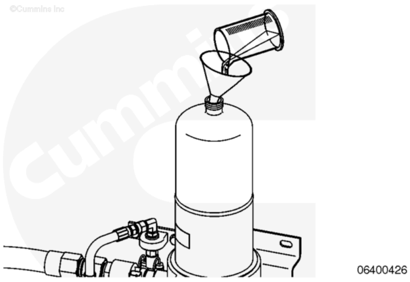
Remove vent cap from top of the can.



Fill filter with clean diesel fuel.

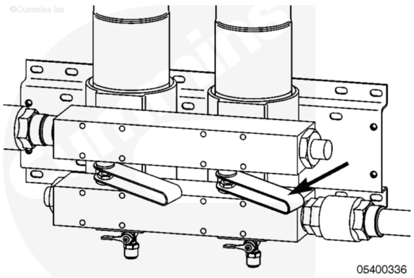
Install the vent cap.

Tighten vent cap by hand **only**.



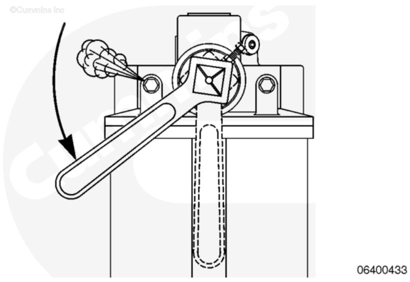
Slowly open valve handle to the ON position, allowing extra time to purge air from the system.

Repeat procedure for the remaining filter replacements.



Open the filter vent and slowly move the selector handle toward center position, allowing fuel to fill the filter that was changed.

Once fuel starts to be discharged from air vent, close the air vent and move the T-handle to center position.

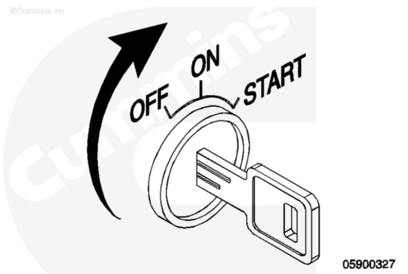


Torque Value:
Air Vent 11 n•m [97 in-lb]

Repeat procedure for the remaining filter, as both filters **must** be used during normal operation.

Marine Applications - Cartridge Filters

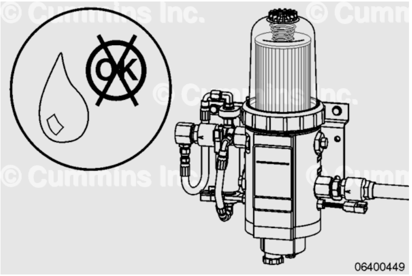
- Prime the fuel system using the electric priming pump.



Finishing Steps

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



All Applications

- Stage 1 filter head **must** be checked for leaks.
- Check plumbing to and from the Stage 1 and Stage 2 filter heads.
- Operate the engine for 1 to 2 minutes and check for leaks.

Last Modified: 13-Jan-2016